MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, RECEIVING

## TYPE 6939

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Twin power pentode, miniature, neutralized, rf amplifier, F1 = 500 MHz

| | | |
|---|---|---|
| Outline | --- | 6-3(EIA) |
| Base | --- | E9-1 |
| Envelope | --- | T6-1/2 |
| Cathode | --- | Coated unipotential |

Base connections:

| Pin No. | --- | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| Element | --- | 2g1 | k, g3 | 1g1 | h | h | 2a | g2 | 1a | hct |

ABSOLUTE-MAXIMUM RATINGS:

| Parameter: | Ef | Eb | Ec1 | Ec2 | Ehk (B) | Ib/a | Ic1/g |
|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | Vdc | Vdc | v | mAdc | mAdc |
| Maximum: | | | | | | | |
| Class C, telep: (intermittent) | 6.9 or 13.9 | 200 | -50 | 200 | 100 | 40 | 4 |
| Class C, telep: (continuous) | 6.9 or 13.9 | 200 | -50 | 200 | 100 | 32 | 3 |
| Class C, teleg: (intermittent) | 6.9 or 13.9 | 250 | -50 | 200 | 100 | 50 | 4 |
| Class C, teleg: (continuous) | 6.9 or 13.9 | 250 | -50 | 200 | 100 | 45 | 3 |
| Minimum: | 5.7 or 11.3 | --- | --- | --- | --- | --- | --- |
| TEST CONDITIONS: | 6.3 or 12.6 | 150 | -3 | 150 | --- | --- | --- |

ABSOLUTE-MAXIMUM RATINGS:

| Parameter: | Pg1/g | Pg2 | Pp/a | Pi/a | TE | Alt |
|---|---|---|---|---|---|---|
| Unit: | W | W | W | W | °C | ft |
| Maximum: | | | | | | |
| Class C, telep: (intermittent) | 0.12 | 2x1.15 | 2.5 | 5.0 | 225 | Note 1 |
| Class C, telep: (continuous) | 0.1 | 2x1.0 | 2.0 | 4.0 | 225 | Note 1 |
| Class C, teleg: (intermittent) | 0.12 | 2x1.75 | 3.75 | 7.0 | 225 | Note 1 |
| Class C, teleg: (continuous) | 0.1 | 2x1.5 | 3.0 | 6.0 | 225 | Note 1 |
| Minimum: | --- | --- | --- | --- | --- | --- |
| TEST CONDITIONS: | --- | --- | --- | --- | --- | --- |

GENERAL:

Qualification - Required

(B) denotes changes



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|---|
| | | | | | | MIN | MAX | |
| | Quality conformance inspection, part 1 | | | | | | | |
| 1301 | Heater current | Ef = 6.3 | 0.65 | II | If | 575 | 625 | mA |
| 1256 | Electrode current (1) (anode) | Ec1 = 0 (see notes 2 and 3) | 0.65 | II | Ib | 33 | 82 | mAdc |
| 1256 | Electrode current (2) (anode) | See note 2 | 0.65 | II | Ib | 7.5 | 31 | mAdc |
| 1256 | Electrode current (3) (anode) | Ec1 = -11.0 Vdc (see note 2) | 0.65 | II | Ib | --- | 125 | µAdc |
| 1266 | Total grid current | Rc1 = 0.1 Meg (see notes 2 and 4) | 0.65 | II | Ic1 | --- | -1.0 | µAdc |
| 1256 | Electrode current (screen) | See note 2 | 0.65 | II | Ic2 | 1.5 | 7.5 | mAdc |
| --- | Pulse cathode current | Eb = Ec2 = 200 Vdc; Du = 0.04; Ef = 5.3 V; egk = +25 v; Rk = 1 ohm; Rl = 0; tp = 800 µs (see note 5) | 0.65 | II | ik | 650 | --- | ma |
| 1201 | Short and discontinuity detection | | 0.4 | II | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | | |
| 1211 | Insulation of electrodes | See notes 6 and 7 | 2.5 | S3 | --- | --- | --- | --- |
| 1336 | Heater-cathode leakage | | --- | --- | Ihk | --- | 40 | µAdc |
| 1306 | Transconductance | Ec1/Ib = 25 mAdc (see note 2) | --- | --- | Sm | 7,800 | 13,200 | µmhos |
| 1331 | Direct-interelectrode capacitance | See note 6 | 6.5 | Code E | Cgp<br>Cin<br>Cout | ---<br>5.1<br>1.4 | 0.2<br>7.7<br>1.8 | pF<br>pF<br>pF |
| 1266 | Primary grid emission (control) | Ef = 7.0 Vac; Pg1 = 0.15 W; Eb = Ec2 = 0 (see notes 6 and 8) | --- | --- | Isg1 | --- | -10 | µAdc |
| 1266 | Primary grid emission (screen) | Ef = 7.0 Vac; Pg2 = 3.5 W; Eb = Ec1 = 0 (see note 8) | --- | --- | Isg2 | --- | -200 | µAdc |
| 1236 | Power oscillation (1) | Push-pull class C amplifier; F = 470 to 500 MHz; Eb = Ec2 = 180 Vdc; Ic1 = 0.75 mAdc/grid; Rc1 = 27,000 ohms/grid; Rc2 = 100 ohms; Ib = 65 mAdc (max); Ic2 = 15 mAdc (max) | --- | --- | Po | 4.1 | --- | W (useful) |
| 1236 | Power oscillation (2) | Power oscillation (1), except Ef = 5.3 Vac | --- | --- | ΔPo Ef | --- | 15 | % |



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 2 -Continued | | | | | | | |
| Ⓑ 1041 | Shock | 450 G; Ef = 6.3 Vac; no other voltages | 15 | S4 | --- | --- | --- | --- |
| 1031 | Vibration fatigue | Ef = 6.3 Vac; F = 50 Hz; no other voltages | 6.5 | Note 9 | --- | --- | --- | --- |
| --- | Post-shock and vibration-fatigue test end points | Insulation of electrodes (see notes 6 and 7) | --- | --- | R | 10 | --- | Meg |
| | | Heater-cathode leakage | --- | --- | Ihk | --- | 70 | μAdc |
| | | Power oscillation (1) | --- | --- | Po | 4.0 | --- | W (useful) |
| | | Pulse cathode current | --- | --- | ik | 600 | --- | ma |
| 1105 | Permanence of marking | | --- | --- | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | | |
| --- | Life test (1) | Group C; power oscillation (1); Ef = 7.8 Vac; Eb = 200 Vdc (see note 10) | --- | --- | --- | --- | --- | --- |
| --- | Life-test end points (1) (100 hours) | Power oscillation (1) | --- | --- | Po | 3.5 | --- | W (useful) |
| | | Power oscillation (2) | --- | --- | ΔPo/Ef | --- | 15 | % |
| | | Pulse cathode current | --- | --- | ik | 450 | --- | ma |
| | | Total grid current | --- | --- | Ic1 | --- | -1 | μAdc |
| | | Primary grid emission (control) | --- | --- | Isg1 | --- | -50 | μAdc |
| | | Primary grid emission (screen) | --- | --- | Isg2 | --- | -200 | μAdc |
| --- | Life test (2) | Group C; power oscillation (1) | --- | --- | --- | --- | --- | --- |
| --- | Life-test end points (2) (1,000 hours) | Power oscillation (1) | --- | --- | Po | 3.5 | --- | W (useful) |
| | | Power oscillation (2) | --- | --- | ΔPo/Ef | --- | 15 | % |
| | | Pulse cathode current | --- | --- | ik | 450 | --- | ma |
| | | Total grid current | --- | --- | Ic1 | --- | -1 | μAdc |
| | | Primary grid emission (control) | --- | --- | Isg1 | --- | -50 | μAdc |
| | | Primary grid emission (screen) | --- | --- | Isg2 | --- | -200 | μAdc |

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. Each unit shall be read separately. Control grid of unit not under test shall be connected to -20 Vdc.
3. The duration of the test shall not exceed 5 seconds.
4. This test to be performed at the conclusion of the holding period.



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NOTES: -Continued

(B) 5. Test both units in parallel. Ec1 shall be sufficient to cutoff the tube. The positive portion of the grid pulse shall be a rectangular wave meeting the pulse characteristic requirements of method 1296. In addition, the maximum amplitude shall occur within the first 20 percent of tp. The pulse shall be applied to the grid by means of a driving circuit which produces the specified peak pulse voltage directly at the grid terminal with respect to cathode. Grid resistance, not exceeding 50-ohms may be inserted to prevent oscillation, provided readjustment of grid drive is made to maintain the specified pulse amplitude directly at the grid terminal. Peak currents shall be measured by a high impedance oscilloscope, or equivalent device, connected across the 1-ohm ±1 percent cathode resistor. The specified limit refers to the maximum pulse amplitude.

6. Each unit shall be tested separately.

7. Voltage shall be applied in series with a 10 megohm resistor.

8. Primary grid-emission tests shall be the last tests performed on the sample selected for quality conformance inspection, part 2. Primary grid emission (control) shall be read prior to primary grid emission (screen).

(B) 9. The vibration-fatigue test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

10. Anode and screen voltage shall be applied intermittently, 2 minutes "on", 8 minutes "off."

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI
Air Force - 11, 17
DSA - ES

User activities:
Army - WC
Navy - AS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2522)



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NAVAL ELECTRONIC SYSTEMS COMMAND
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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-35264

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